Fall 2026 DSC 242: Homework 3

Due 11:59PM Friday, October 23

Submission. Problems 1 and 2 are written problems. Problem 3 is an experimental problem: submit a single PDF report (for example, exported from a Jupyter notebook) that contains your figures, a short discussion for each part, and the code that produced the figures (as notebook cells or as an appendix). Every figure should have labeled axes, a legend, and a caption stating the parameters and the number of trials.

1. (Vershynin 6.4.6, 10 points) Let $X_1, \dots, X_n$ be independent subgaussian random variables with $\mathbb{E}[X_i] = 0$, and let $\epsilon_1, \dots, \epsilon_n$ be i.i.d. Rademacher random variables independent of $X_1, \dots, X_n$. Show that for universal constants $C, c > 0$,

$$c\left\|\sum_{i=1}^n \epsilon_i X_i\right\|_{\psi_2} \le \left\|\sum_{i=1}^n X_i\right\|_{\psi_2} \le C\left\|\sum_{i=1}^n \epsilon_i X_i\right\|_{\psi_2}.$$

   (Hint: Use the Rademacher symmetrization lemma from Lecture 6.)

2. (Dantzig selector and LASSO, 40 points) Consider the linear model $y = X\beta^* + \epsilon$, where the design matrix $X \in \mathbb{R}^{n\times p}$, unknown signal $\beta^* \in \mathbb{R}^p$ and noise $\epsilon \in \mathbb{R}^n$. Suppose $\|\beta^*\|_0 = \kappa$ and denote $S = \mathrm{supp}(\beta^*)$ as the support of β^*. Suppose $\|\frac{1}{n}X^\top \epsilon\|_\infty \le \lambda$, and X satisfies the Restricted Eigenvalue (RE) condition, i.e., for any $r \in \mathbb{R}^p$ such that $\|r_{S^c}\|_1 \le 3\|r_S\|_1$, it holds that $\frac{1}{n}\|Xr\|_2^2 \ge c\|r\|_2^2$ for some constant $c > 0$.

   The Dantzig selector is defined as

$$\hat{\beta}_D = \operatorname*{argmin}_{\beta\in\mathbb{R}^p} \|\beta\|_1 \quad \text{s.t.} \quad \left\|\frac{1}{n}X^\top(y - X\beta)\right\|_\infty \le \lambda. \tag{0.1}$$

   (a) (Basic inequality, 10 points) Show that β^* is a feasible solution to the optimization problem in (0.1), and thus $\|\hat{\beta}_D\|_1 \le \|\beta^*\|_1$.

   (b) (Cone condition, 10 points) Show that $\hat{\beta}_D$ satisfies the cone condition $\|(\hat{\beta}_D - \beta^*)_{S^c}\|_1 \le \|(\hat{\beta}_D - \beta^*)_S\|_1$, and thus $\|\hat{\beta}_D - \beta^*\|_1 \le 2\sqrt{\kappa}\|\hat{\beta}_D - \beta^*\|_2$.

   (c) (10 points) Show that $\|\frac{1}{n}X^\top X(\hat{\beta}_D - \beta^*)\|_\infty \le 2\lambda$ by using the constraint of (0.1), and then show that $\frac{1}{n}\|X(\hat{\beta}_D - \beta^*)\|_2^2 \le 2\lambda\|\hat{\beta}_D - \beta^*\|_1$.

   (d) (10 points) Finally, show that $\|\hat{\beta}_D - \beta^*\|_1 \le C\kappa\lambda$ for some constant $C > 0$, that is, the Dantzig selector achieves the same error rate as LASSO.

3. (Sparse linear regression in practice, 50 points) In this problem you will check, by simulation, the predictions of the LASSO theory from Lectures 7–8. We recommend Python: `sklearn.linear_model.Lasso` solves the LASSO, and `cvxpy` (or `scipy.optimize.linprog`) solves the linear program in part (b). Throughout, generate data as follows:

   - design $X \in \mathbb{R}^{n\times p}$ with i.i.d. $\mathcal{N}(0, 1)$ entries;

- signal $\beta^* \in \mathbb{R}^p$ supported on a uniformly random set S of size k, with $\beta^*_j \in \{-1, +1\}$ uniformly at random for $j \in S$;
- noise $\epsilon \sim \mathcal{N}(0, \sigma^2 I_n)$ with $\sigma = 1$ (except in part (b), where $\sigma = 0$);
- response $y = X\beta^* + \epsilon$.

The LASSO estimator is

$$\hat{\beta}_\lambda = \operatorname*{argmin}_{\beta \in \mathbb{R}^p} \frac{1}{2n}\|y - X\beta\|_2^2 + \lambda\|\beta\|_1.$$

(Note the $\frac{1}{2n}$ in front of the squared loss. This is the convention of `sklearn`, where the above is `Lasso(alpha=λ, fit_intercept=False)`; other libraries may normalize differently, so make sure that the λ you pass corresponds to this objective.) Define the reference value

$$\lambda_0 = \sigma\sqrt{\frac{2\log p}{n}},$$

and in parts (c)–(e) use $\lambda = \frac{1}{2}\lambda_0$ unless stated otherwise. Every reported quantity is an average over independent trials (fresh X, β^*, ϵ in each trial); use at least 20 trials per configuration, and more where it is cheap. With the sizes suggested below, each part runs in a few minutes on a laptop; if you need to reduce the grids or the number of trials, say so in the report.

(a) (The noise level, 5 points) Let $p = 1000$ and $n \in \{50, 100, 200, 400, 800, 1600\}$. Estimate $\mathbb{E}\|\frac{1}{n}X^\top\epsilon\|_\infty$ from at least 100 trials. Plot it against n on log-log axes together with the curve $\lambda_0 = \sigma\sqrt{2\log p/n}$, and plot the ratio $\mathbb{E}\|\frac{1}{n}X^\top\epsilon\|_\infty/\lambda_0$ against n (see Figure 1 for the format). What is the slope of the log-log plot, and is the ratio constant? Explain the value of the ratio using the union bound from Lecture 7: is the constant $\sqrt{2}$ in λ_0 sharp?

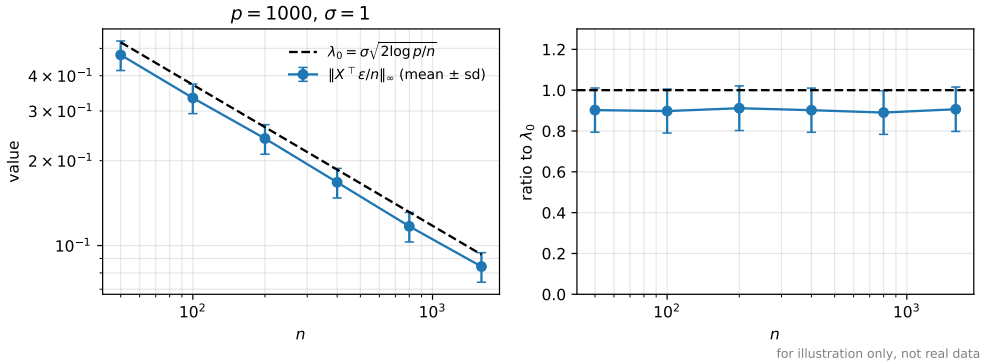


Figure 1: Expected format of the figure for part (a), for illustration only (not real data). Left: $\mathbb{E}\|\frac{1}{n}X^\top\epsilon\|_\infty$ and λ_0 against n. Right: their ratio.

(b) (Noiseless phase transition, 20 points) Set $\sigma = 0$ and $p = 256$. For each $n \in \{8, 16, \ldots, 256\}$ and $k \in \{1, 3, 5, \ldots, 63\}$, solve the *basis pursuit* problem

$$\hat{\beta}_{\mathrm{BP}} = \operatorname*{argmin}_{\beta \in \mathbb{R}^p} \|\beta\|_1 \quad \text{s.t.} \quad X\beta = y,$$

which is the $\lambda \to 0$ limit of the LASSO, and record whether the recovery is exact, $\|\hat{\beta}_{\mathrm{BP}} - \beta^*\|_2 \le 10^{-4}\|\beta^*\|_2$. (Basis pursuit is a linear program: write $\beta = \beta^+ - \beta^-$ with $\beta^\pm \ge 0$.) Display the empirical success probability as a heat map over the (k, n) grid, with k on the horizontal axis and n on the vertical axis, as in Figure 2. Then:

i. For each k, let $n_{50}(k)$ be the smallest n at which the success rate reaches $1/2$. Fit a constant C such that $n_{50}(k) \approx C\, k \log(ep/k)$ (for example by least squares), and overlay the curve $n = C\, k \log(ep/k)$ on the heat map. Does the shape of the empirical boundary follow $k \log(ep/k)$? What is your C?

ii. Lecture 8 shows that a Gaussian design satisfies the restricted eigenvalue condition once $n \geq C' k \log(ep/k)$, which implies exact recovery. Explain why this result predicts the *shape* of the boundary but not its *location*, and why the transition in your heat map is sharp (a narrow band from success rate 0 to 1) rather than gradual.

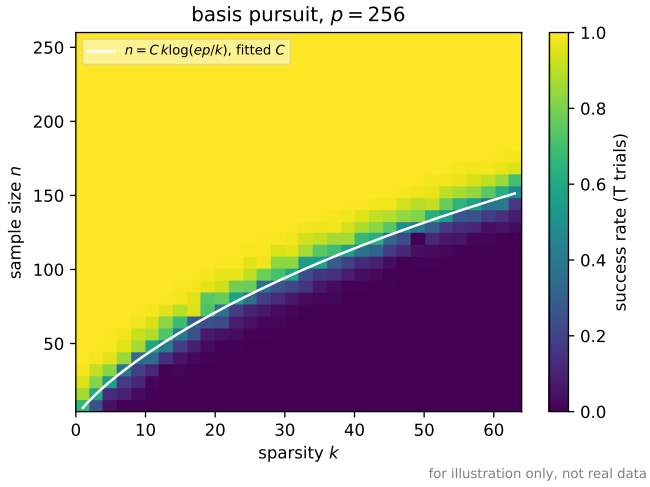


Figure 2: Expected format of the heat map for part (b), for illustration only (not real data). Yellow is success rate 1, dark blue is 0. The white curve is $n = C\, k \log(ep/k)$ with your fitted C.

(c) (Error rates in n and k, 10 points) Set $p = 512$ and $\lambda = \frac{1}{2}\lambda_0$.

i. Fix $k = 10$ and let $n \in \{200, 400, 800, 1600, 3200\}$. Plot the average squared error $\mathbb{E}\|\hat{\beta}_\lambda - \beta^*\|_2^2$ against n on log-log axes. Fit the slope and compare it with the theoretical rate.

ii. Fix $n = 800$ and let $k \in \{5, 10, 15, 20, 25, 30\}$. Plot $\mathbb{E}\|\hat{\beta}_\lambda - \beta^*\|_2^2$ against k on linear axes. Is the dependence linear in k?

See Figure 3 for the format.

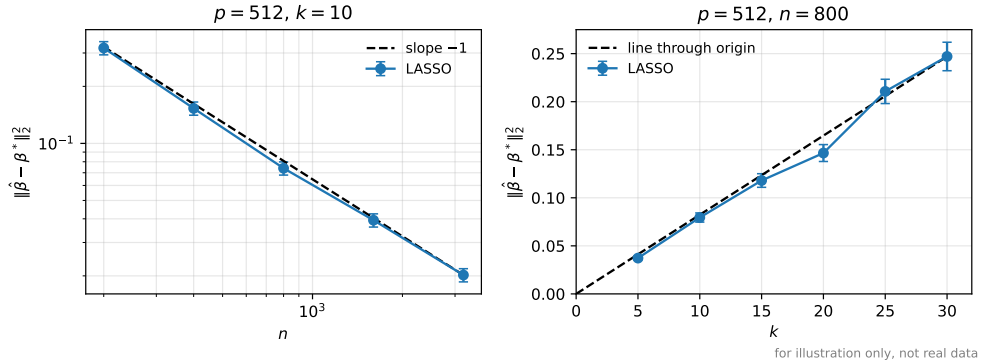


Figure 3: Expected format of the figures for part (c), for illustration only (not real data). Error bars are one standard error of the mean.

(d) (Rescaled sample size, 10 points) Consider the four configurations

$$(p, k) \in \{(128, 12),\ (256, 16),\ (512, 23),\ (1024, 32)\}$$

(so $k \approx \sqrt{p}$), and for each let $n = r \cdot k \log p$ for $r \in \{1, 2, 3, 4, 6, 8, 12, 16\}$ (rounded to an integer). Make two plots side by side (see Figure 4): the average squared error $\mathbb{E}\|\hat{\beta}_\lambda - \beta^*\|_2^2$ against n for the four configurations, and the *rescaled* error $\mathbb{E}\|\hat{\beta}_\lambda - \beta^*\|_2^2/k$ against the *rescaled sample size* $n/(k \log p)$. Explain, using the bound from Lecture 8, why the four curves should approximately collapse onto a single curve in the second plot but not in the first. What does this say about how the ambient dimension p enters the problem?

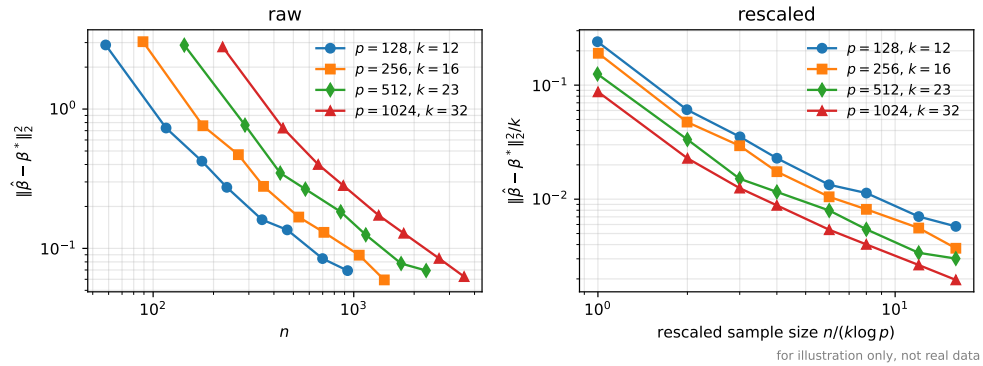


Figure 4: Expected format of the figures for part (d), for illustration only (not real data).

(e) (Choice of λ, 5 points) Set $p = 512$, $k = 10$ and $n \in \{200, 400, 800\}$. For $\lambda = c\lambda_0$ with c on a geometric grid from 0.03 to 3 (about 10 values), plot $\mathbb{E}\|\hat{\beta}_\lambda - \beta^*\|_2^2$ against c on log-log axes, one curve per n, and mark $c = 1$ (see Figure 5). Where is the minimum? Explain what goes wrong when λ is too small and when it is too large. The proof in Lecture 7 requires $\lambda \geq 2\|\frac{1}{n} X^\top \epsilon\|_\infty$; compare this with the empirically optimal λ in view of part (a), and comment.

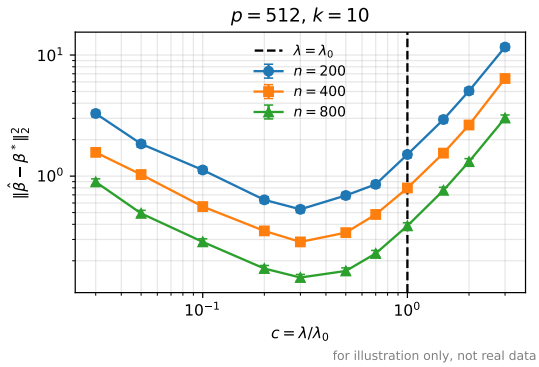


Figure 5: Expected format of the figure for part (e), for illustration only (not real data).